Parent Messaging (Room Chat)- Admin / Staff's view

0) Scope & Assumptions

- **Users:** Org Admins, School Admins, Teachers/ Staff
- **Purpose:** Enable secure, role-based communication between staff and parents/guardians.
- **Tenancy:** Messaging is scoped to the selected organization and school context.
- **Session:** Browser-based SPA; JWT or session cookie.
- **Timezone:** Use school’s timezone for timestamps.

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1) User Journey

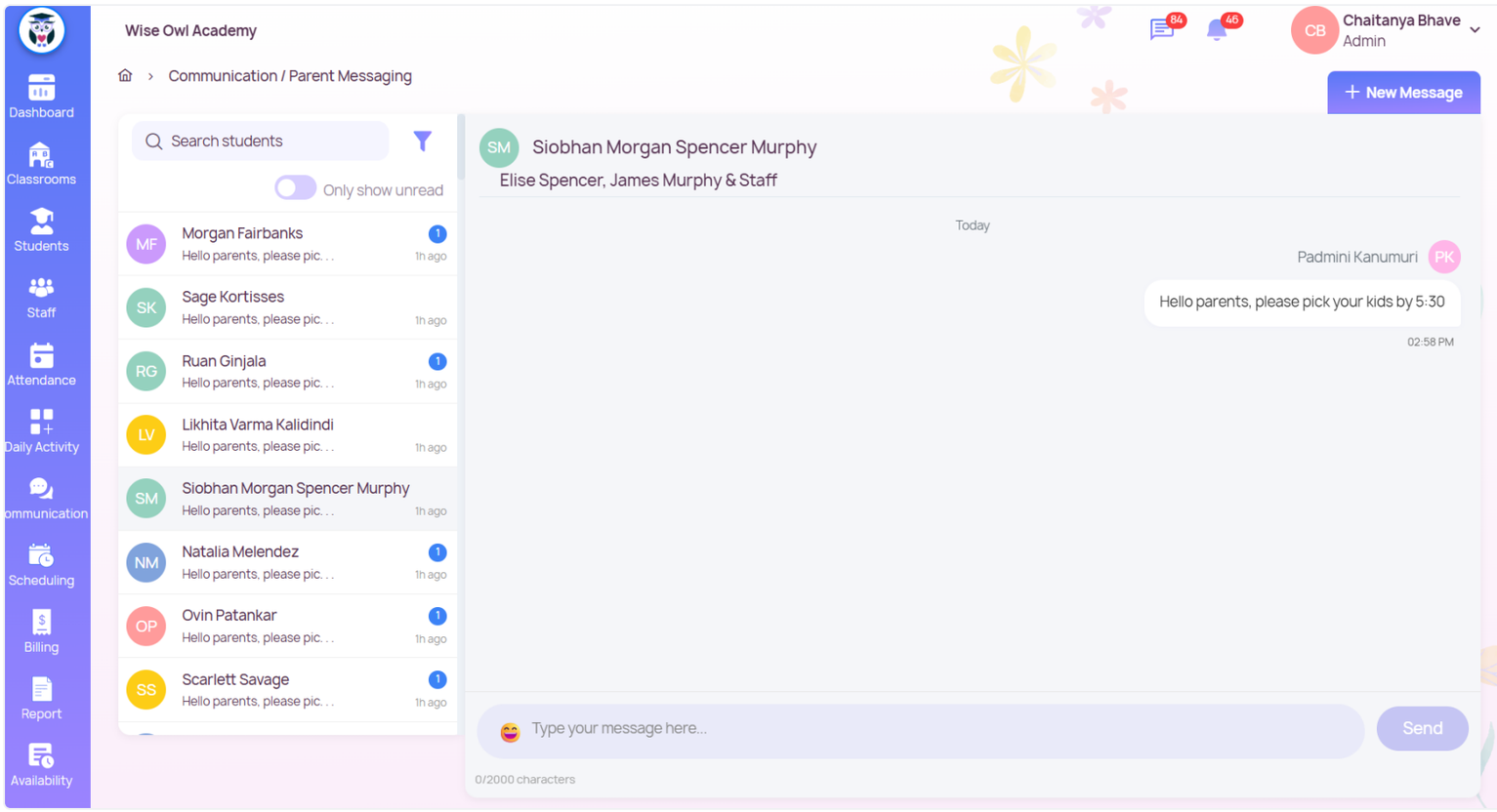
Persona: Teacher/Admin
Actions:

- Navigate to **Communication → Parent Messaging→ Room Chats**
- View inbox with conversation threads.
- Compose a new message (bulk) OR send message in a single chat.
- Track delivery and read status.

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2) Core Functionalities

2.1 Inbox & Conversation View



- Sidebar lists conversations:
  - Student name, Photo/ initials + last message preview with timestamp
  - Unread badge count.
- Search student (Active & upcoming students): Should persist until changed
- Room Chat for a student will include following participants when created (All Admins+ directors + Allowed staff + All parents) (edits /deletes will cascade & update participants

Field Name	Data Type	Default Value	Validations	Additional Notes
Search student		chronological order.		
Classroom Filter	Dropdown Multiselect	Empty (showing all students in all classes) for staff primary classroom	Must match active classrooms	Multi-select; displayed in filter panel
Tag Filter		Empty (showing all students)	Must match predefined tags	Multi-select; displayed in filter panel
Status Filter	Enum	Active	Allowed values:	Default: Active; Allow

			Active, Upcoming	upcoming & Inactive students only if filter applied
clear all	Button			Clears all filter (no need to click apply again)

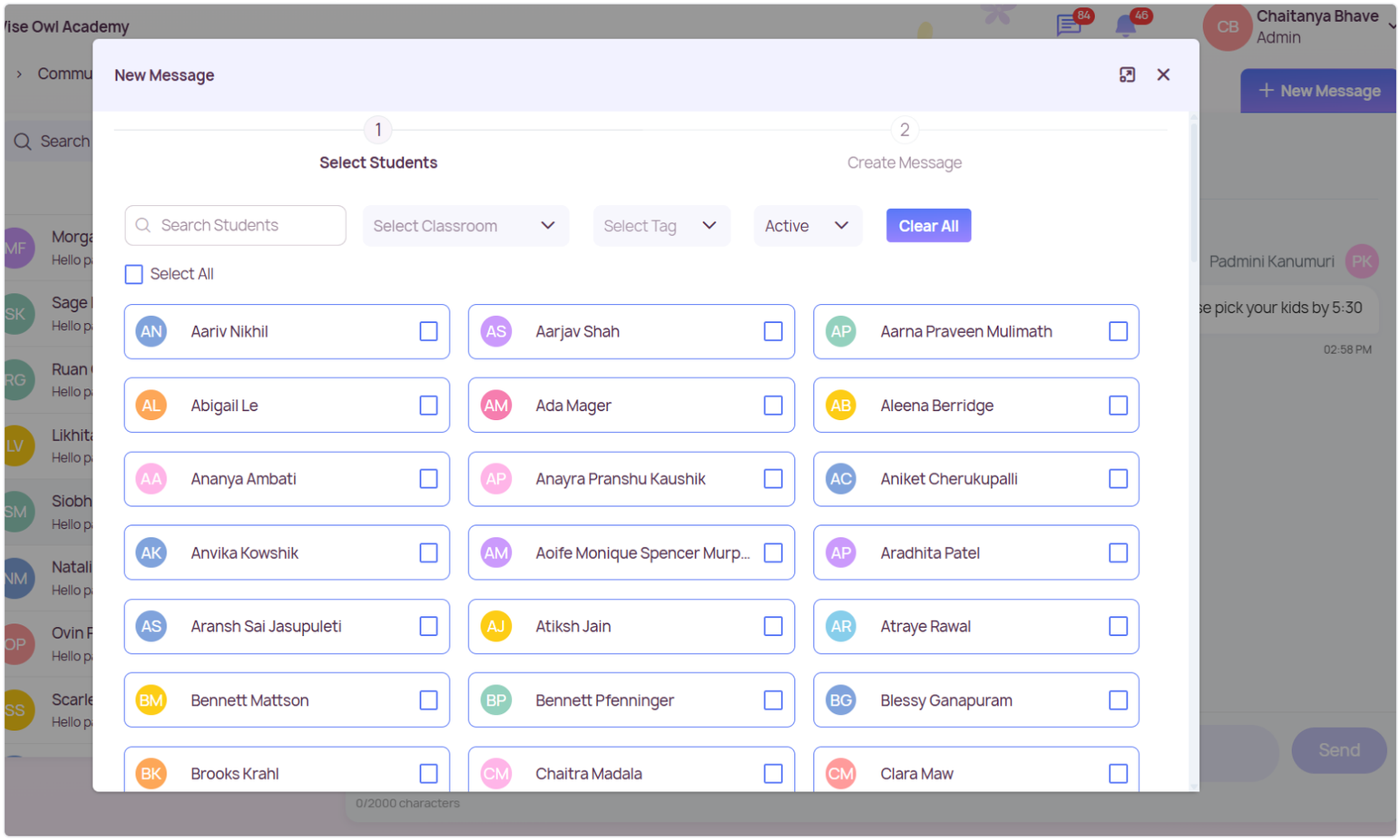
- Toggle: “Show only unread.” (filters only conversations with at least one unread message)
- Clicking a conversation opens threaded view:
  - Messages in chronological order.
  - Each message shows sender name, timestamp.
  - **Compose & send message :**Inside a conversation thread, user can compose and send a message **specific to that chat** (single recipient context).

Field Name	**Data Type**	**Default Value**	**Validations**	**Additional Notes**
Message Text	String	Empty	Required; Max 2000 characters	Supports **Rich Text** (bold, italic, underline, line spacing when enter etc.) , Emojis & URLs
Character Counter	Integer	6000	Decrements as user types	Displayed below text box
Attachement	file	None	optional, JPEG, PNG, HEIC, PDF & Docs files allowed each file<10mb, Multiple allowed	Photo visible directly in chat. , PDF or doc is like file (sameas in whatsapp) multiple photo or file view same as whatsapp
Send Button	Boolean	Disabled	Enabled only if message	Primary CTA

			text is not empty	

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2.2 Compose New Message (Bulk Action)



a. **Flow:** Click **New Message**.

b. **Step 1:** Select recipients:
   - Search bar for student names.
   - Filters: Classroom, Tags, Status (Active/Inactive).
   - **Default:** Show only active students; exclude inactive; allow upcoming students.
   - “Select All” option.
   - Ability to remove selected recipients.

Field Name	Data Type	Default Value	Validations	Additional Notes
Search Student	String	Empty	Max length: 100 chars	Supports partial match; case-insensitive search

Classroom Filter	Dropdown Multiselect	Empty (showing all students in all classes)	Must match active classrooms	Multi-select; displayed in filter panel
Tag Filter		Empty (showing all students)	Must match predefined tags	Multi-select; displayed in filter panel
Status Filter	Enum	Active	Allowed values: Active, Upcoming	Default: Active; Inactive students excluded
Student List	Array, link object with photo /initials	Active students only	At least one student must be selectable to go to next step	Upcoming students can appear if filter selected for that; inactive hidden
Clear all				clears all the filters to defaults
Select All Checkbox	Boolean	False	N/A	Selects all currently filtered students
Selected Recipients	Array	Empty	At least one recipient required before sending	Ability to remove recipients individually

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Step 2 – Message Composition

- Rich text editor (bold, italic, underline) with emojis and optional attachment.
- URL-compatible text (clickable links).
- Character counter (max 6000 chars).
- **Send** button enabled only if:

- At least one recipient selected.
- Message text is not empty.

Field Name	Data Type	Default Value	Validations	Additional Notes
Message Text	String	Empty	Required; Max 2000 characters	Supports **Rich Text** (bold, italic, underline, line spacing when enter etc.) , Emojis & URLs
Character Counter	Integer	6000	Decrements as user types	Displayed below text box
Attachement	file	None	optional, JPEG, PNG, HEIC, PDF & Docs files allowed <10mb	Photo visible directly in chat. , PDF or doc is like file (sameas in whatsapp)
Go Back	Boolean	Active		Goes back to student selection
Send Button	Boolean	Disabled	Enabled only if message text is not empty AND ≥1 recipient selected	Primary CTA

- **Rich Text & URL support** in message composer.
- **Default filter behavior:** Active students only; exclude inactive; allow upcoming.
- **Validation:** Send button logic tied to both steps.

2.3 Message Actions

- On sent messages:
  - **Copy** text.

    Edit message.

    - **Delete** message.
    - **Info / Status** → Opens “Message Status” popup:
        - Original message content.
        - Delivered To: list of recipients.
        - Read By: list of recipients who read the message.
- Read receipts visible per message.

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3) Filter Functionality

- Filter icon opens panel with:
    - Classroom filter (multi-select).
    - Tag filter (multi-select).
    - Status filter (Active, Upcoming).
    - Search filter for student names.
- Default: Active students only.

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4) Field Specifications

Field Name	Data Type	Default Value	Validations	Notes
Message Text	String	Empty	Max 2000 chars	Rich text + URL support
Recipients	Array	Empty	At least 1	Selected via filters/search
Filters	Enum	Active only	Classroom/Tag/Status	Optional

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5) Acceptance Criteria

- **AC1:** Inbox shows all conversations sorted by latest message.
- **AC2:** Unread badge count updates in real-time.
- **AC3:** “Send” disabled until message text and recipient(s) are provided. & once pressed disabled again until action is completed.
- **AC4:** Delivery status accurate within 2 seconds of sending.
- **AC5:** Read receipts update within 5 seconds after recipient views message.
- **AC6:** Filters return correct student list; default excludes inactive.
- **AC7:** Character counter decrements as user types.

  AC8: Rich text formatting and clickable URLs supported in composer.

- **AC9:** Single-chat composer sends message only to that conversation’s recipient.
- **AC10:** Delete action shows deleted message in thread (soft delete with text " This message has been deleted").
- AC 11: All messages from parent will be visible to staff on left side in Pink gradient background. & all other staff's messages on right side in white background.
- Ac12: All messages from staff will be visible to parents on left side. all other parent's messages on right side in white background.
- AC13: Adding/ editing /removing classroom of the students should update chat participants accordingly (Admins+ allowed staff +all parents)
- AC14: Adding/ editing /removing classroom of the staff should update chat participants accordingly (Admins+ allowed staff +all parents)
- Ac15: Adding/ editing /removing parents of the students should update chat participants accordingly (Admins+ allowed staff +all parents)
- AC16: All Cross platform actions (web & mobile app) Should perform exactly the same.
- AC17: Alerts & notifications should be as per alert sheet `https://docs.google.com/spreadsheets/d/1_oBkliWUoZ_aX-OeDs0f_HfbRcym9fAEephuqanm9WM/edit?pli=1&gid=0#gid=0` & Private (https://app.clickup.com/9017410820/docs/8cqnh84-10057/8cqnh84-9217)
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6) Security Requirements

- Role-based access: All admins, Only authorized staff (staff allowed for the classroom of that student) can send messages.
- Validate recipient IDs server-side.
- Audit logs: userId, messageId, timestamp, orgId, siteId, action (send/edit/delete).

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7) Accessibility & Localization

- Keyboard navigation for all fields.
- ARIA labels for message status popup.
- WCAG 2.1 AA compliance.
- All strings i18n-ready.

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8) Non-Functional Requirements

- **Performance:** Inbox load < 500 ms; message send < 800 ms.
- **Scalability:** Support up to 10K conversations per user without UI lag.
- **Reliability:** 99.9% uptime for messaging service.

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9) QA Test Matrix

- **Happy Path:** Compose → Send → Delivery → message actions (copy, edit,Read status, delete).
- **Edge Cases:** Empty message, no recipients, API failure.
- **Filters:** Classroom/tag/status combinations.
- **Unread toggle:** Shows only unread threads.
- **Rich text:** Bold, italic, links render correctly.

More **Acceptance Criteria**

1. **Inbox Functionality**
   - AC1: Inbox displays all conversations sorted by the latest message timestamp.
   - AC2: Unread badge count updates in real-time when a new message is received.
   - AC3: Search functionality persists until manually cleared or modified.
   - AC4: Filters (Classroom, Tags, Status) return accurate student lists based on selected criteria.
   - AC5: "Show only unread" toggle filters conversations with at least one unread message.
   - AC6: Clicking a conversation opens the threaded view with messages in chronological order.
2. **Message Composition (Single Chat)**
   - AC7: "Send" button is disabled until the message text is not empty.
   - AC8: Message composer supports rich text formatting (bold, italic, underline), emojis, and URLs.
   - AC9: Character counter decrements as the user types, with a maximum limit of 2000 characters.
   - AC10: Attachments (JPEG, PNG, HEIC, PDF, DOC) are supported, with each file <10MB and multiple files allowed.
   - AC11: Sent messages appear in the conversation thread immediately after sending.
   - AC12: Delivery status updates within 2 seconds of sending.
   - AC13: Read receipts update within 5 seconds of the recipient viewing the message.
3. **Bulk Message Composition**
   - AC14: Default filter behavior shows only active students; excludes inactive students unless explicitly filtered.
   - AC15: At least one recipient must be selected to enable the "Send" button.
   - AC16: "Select All" option selects all students based on the applied filters.
   - AC17: Recipients can be removed individually from the selected list.
   - AC18: Rich text formatting, emojis, URLs, and attachments are supported in bulk messages.

- AC19: "Go Back" button navigates back to recipient selection without losing progress.

4. **Message Actions**
   - AC20: Sent messages can be copied, edited, or deleted.
   - AC21: Deleted messages are replaced with "This message has been deleted" in the thread (soft delete).
   - AC22: Message status popup displays the original message, delivery status, and read receipts.
5. **Participant Updates**
   - AC23: Adding, editing, or removing classrooms for students updates chat participants accordingly (Admins + allowed staff + all parents).
   - AC24: Adding, editing, or removing classrooms for staff updates chat participants accordingly.
   - AC25: Adding, editing, or removing parents for students updates chat participants accordingly.
6. **Cross-Platform Consistency**
   - AC26: All actions (web and mobile app) perform identically.
7. **Security**
   - AC27: Only authorized staff (allowed for the student's classroom) can send messages.
   - AC28: Recipient IDs are validated server-side.
   - AC29: Audit logs capture userId, messageId, timestamp, orgId, siteId, and actions (send/edit/delete).
8. **Accessibility & Localization**
   - AC30: All fields are keyboard-navigable.
   - AC31: ARIA labels are present for message status popups.
   - AC32: WCAG 2.1 AA compliance is met.
   - AC33: All strings are i18n-ready.
9. **Performance & Reliability**
   - AC34: Inbox loads within 500ms.
   - AC35: Messages are sent within 800ms.
   - AC36: The system supports up to 10K conversations per user without UI lag.
   - AC37: Messaging service maintains 99.9% uptime.

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10) Gherkin Scenarios

```
Feature: Messaging
Scenario: Send message in single chat
  Given I am viewing a conversation thread
  When I type ”Hello” in the composer
  And click Send
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  Then the message appears in that thread only
```

Inbox Functionality

```
Feature: Inbox
Scenario: Display all conversations sorted by latest message
  Given I am logged into the Parent Messaging system
  When I view the inbox
  Then all conversations are displayed sorted by the latest message timestamp

Scenario: Update unread badge count in real-time
  Given I have an unread message in a conversation
  When a new message is received
  Then the unread badge count updates in real-time
```

Message Composition (Single Chat)

```
Feature: Single Chat Messaging
Scenario: Send a message in a single chat
  Given I am viewing a conversation thread
  When I type "Hello" in the composer
  And click "Send"
  Then the message appears in that thread only

Scenario: Disable "Send" button until message text is entered
  Given I am viewing a conversation thread
  When the message composer is empty
  Then the "Send" button is disabled
  When I type "Hi"
  Then the "Send" button is enabled

Scenario: Attach a file to a message
  Given I am composing a message
  When I attach a JPEG file less than 10MB
  Then the file is visible directly in the chat
```

Bulk Message Composition

```
Feature: Bulk Messaging
Scenario: Send a bulk message to active students
  Given I click "New Message"
  When I view the student selection
  Then only active students are shown by default
  When I select "Classroom A" and "Classroom B"
  And type "Important update" in the composer
  And click "Send"
  Then the message is sent to all selected students

Scenario: Remove a recipient from the selected list
  Given I have selected multiple recipients
  When I remove one recipient
  Then the recipient is no longer in the selected list
```

```
Scenario: Bulk message excludes inactive students
  Given I click New Message
  When I view student selection
  Then only active students are shown
```

Message Actions

```
Feature: Message Actions
Scenario: Delete a sent message
  Given I have sent a message
  When I click "Delete"
  Then the message is replaced with "This message has been deleted"

Scenario: View message status
  Given I have sent a message
  When I click "Info / Status"
  Then I see the original message content, delivery status, and read receipts
```

Participant Updates

```
Feature: Participant Management
Scenario: Update participants when a classroom is edited
  Given a student is moved to a new classroom
  When I view the conversation participants
  Then the participants list reflects the updated classroom
```

Cross-Platform Consistency

```
Feature: Cross-Platform Messaging
Scenario: Perform actions on web and mobile app
  Given I send a message on the web app
  When I view the conversation on the mobile app
  Then the message is visible in the same thread
```

Security

```
Feature: Security
Scenario: Restrict messaging to authorized staff
  Given I am a staff member not assigned to a student's classroom
  When I try to send a message to that student
  Then I am not allowed to send the message
```

Performance

```
Feature: Performance
Scenario: Inbox loads within 500ms
  Given I log into the Parent Messaging system
  When I open the inbox
  Then the inbox loads within 500ms
```

```
Feature: Notifications for Messaging

Scenario: Display in-app notification for a new message
  Given I am logged into the Parent Messaging system
  And I have an unread message in a conversation
  When a new message is received
  Then an in-app notification is displayed
  And the notification includes the sender's name and a preview of the message

Scenario: Push notification for a new message
  Given I have enabled push notifications for the Parent Messaging app
  And I have an unread message in a conversation
  When a new message is received
  Then a push notification is sent to my device
  And the notification includes the sender's name and a preview of the message

Scenario: Notification clears when the message is read
  Given I have an in-app or push notification for a new message
  When I open the conversation thread
  Then the notification is cleared

Scenario: Notification for bulk messages
  Given I am a recipient of a bulk message
  When the message is sent
  Then I receive an in-app notification
  And I receive a push notification if enabled

Scenario: Notification for message delivery failure
  Given I have sent a message
  And the message fails to deliver
  When the system detects the failure
  Then I receive an in-app notification about the delivery failure

Scenario: Notification for message read status
  Given I have sent a message
  And the recipient reads the message
  When the system updates the read status
  Then I receive an in-app notification that the message was read

Scenario: Notification settings update
  Given I am logged into the Parent Messaging system
  When I update my notification preferences
  Then the system saves my preferences
  And notifications are sent according to the updated settings
```